Te Papa's COLOSSAL

An unexpected catch

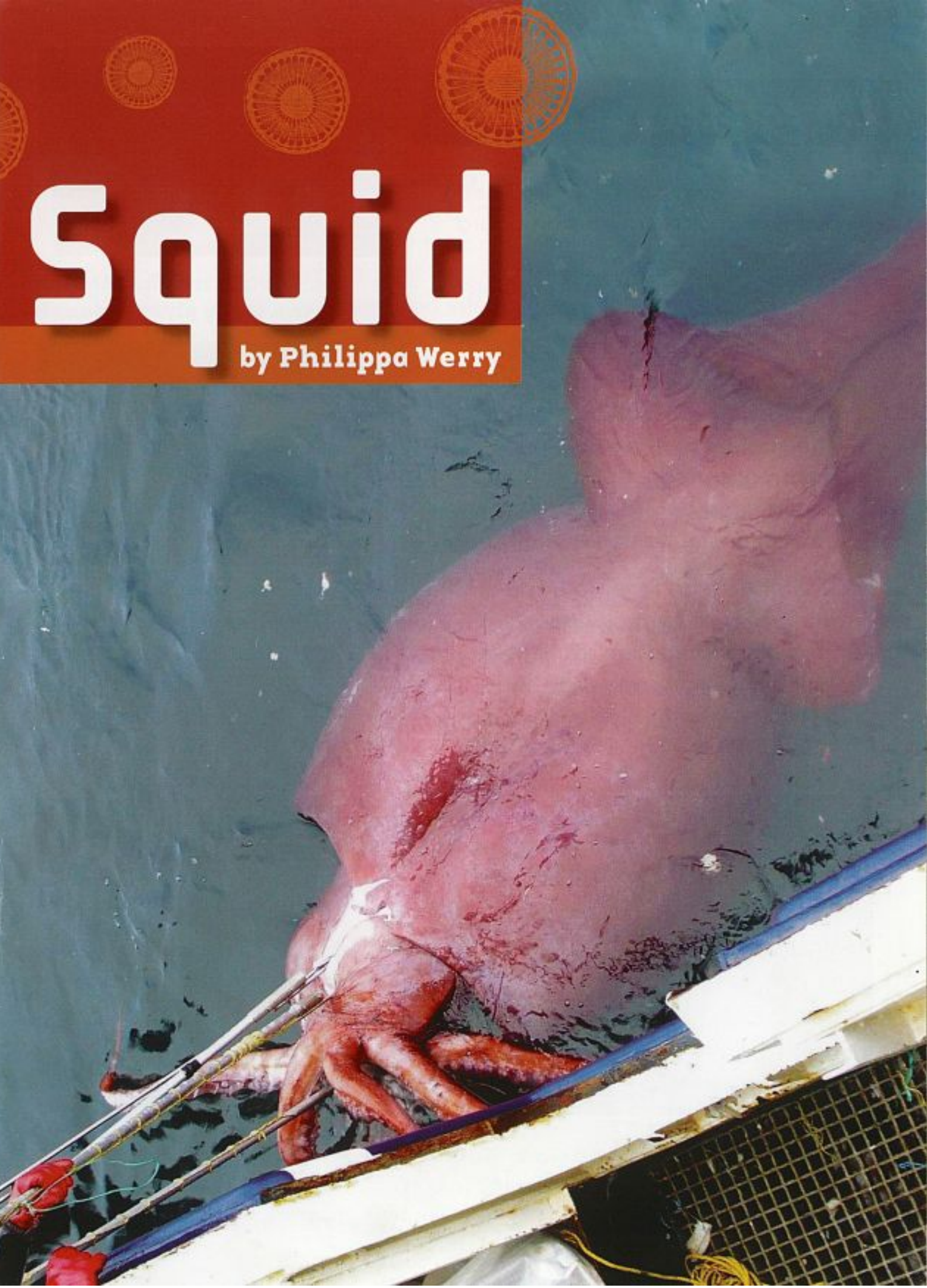
In February 2007, a New Zealand fishing boat called the *San Aspiring* was fishing for Antarctic toothfish in the Ross Sea. But something else was hunting for toothfish as well. When the crew pulled up their longlines, they were astonished to see what came to the surface. An enormous, orange-red, deep-sea monster was wallowing alongside the boat.

It was a colossal squid, still hanging onto a toothfish, and it was only just alive. The crew realised this was an opportunity to learn more about these mysterious creatures. It took them two hours to heave the squid on board and store it in the boat's freezer. When the boat returned to port, the squid was put in a cool store in Timaru. Later the Ministry of Fisheries donated it to Te Papa. It was taken to Wellington and kept in a walk-in freezer in the research laboratories while the scientists decided what to do with it.

This was the first complete adult colossal squid ever landed. Scientists from all round the world wanted to see it. But there were some big problems to be worked out first.

Squid

by Philippa Werry



A colossal headache

When the colossal squid had been bundled into the freezer on board the *San Aspiring*, it froze into a solid block. The biggest problem was how to make sure that the outside tentacles didn't defrost too quickly and start to rot while thicker body parts were still frozen. The squid was folded up in layers like an accordion, but it couldn't be unfolded before it was properly thawed or the tissues would break.

At the same time, Te Papa staff wanted to let as many people as possible share in the excitement of the discovery, and they had to decide how to display the squid after they had thawed and examined it.

The big defrost

It takes several hours for a 1.5-kilogram frozen chicken to defrost for your dinner. So how long would it take for a 495-kilogram frozen colossal squid to defrost?

1

Building the thawing tank using plywood panels

2

The tank before it is sealed and lined

3

The tank sealed and lined

4

The tank, fully lined with plastic, being tested by filling it with water

The plan

Nobody was sure exactly how long the squid was, so scientists had to guess how long to make a tank to hold it.
A tank measuring 6.5 metres by 2 metres was built. It was lined with plastic and filled with 10 000 litres of water. Then 300 litres of salt were dissolved into it so that the squid would be surrounded by water that was similar to sea water. Salt also depresses or lowers the freezing point of water, so the ice would melt while the squid stayed at about 0°C.

A replica block of ice the same weight as the squid (495 kilograms) was frozen in a plastic container and then thawed. This helped the scientists to estimate how long the thawing process might take. It also let them test out the crane and lifting gear that would be used to move the colossal squid.

Scientists from all round the world gathered to be part of the event. Nothing like this had ever been done before, so Te Papa's scientists had to be ready to change their plans at a moment's notice if things didn't work out as expected.

The plastic container, filled with water and frozen, being cut open to remove the ice so that experiments with lifting equipment and melting rates can be carried out.

Monday, 28 April 2008

The colossal squid is taken out of the freezer at 3 p.m. Two overhead hoists are used to lower it into the tank. The tarpaulin and ropes are removed, and it is left to slowly thaw over the next two days.

Tuesday, 29 April

It is a warm day, and the water temperature is creeping up. A tonne of crushed ice from a local fishing company is added to the tank to keep it cold enough.

The squid is still wrapped in the net it was caught in. Scientists get into the tank and carefully cut the net away. They wear gumboots and waders, but the water is very cold and their feet soon go numb. They can't wear gloves because they need to be able to feel the squid, so their fingers are very cold as well.

Lowering the squid into the tank

The frozen squid in the net

Wednesday, 30 April

The colossal squid is still not completely thawed. Bubble wrap and floating bags are used to protect the tentacles and stop them from breaking off. The scientists have to constantly monitor progress. They run water over the squid all morning. At one stage, they pump in some warm water – being extra careful not to cook the squid!

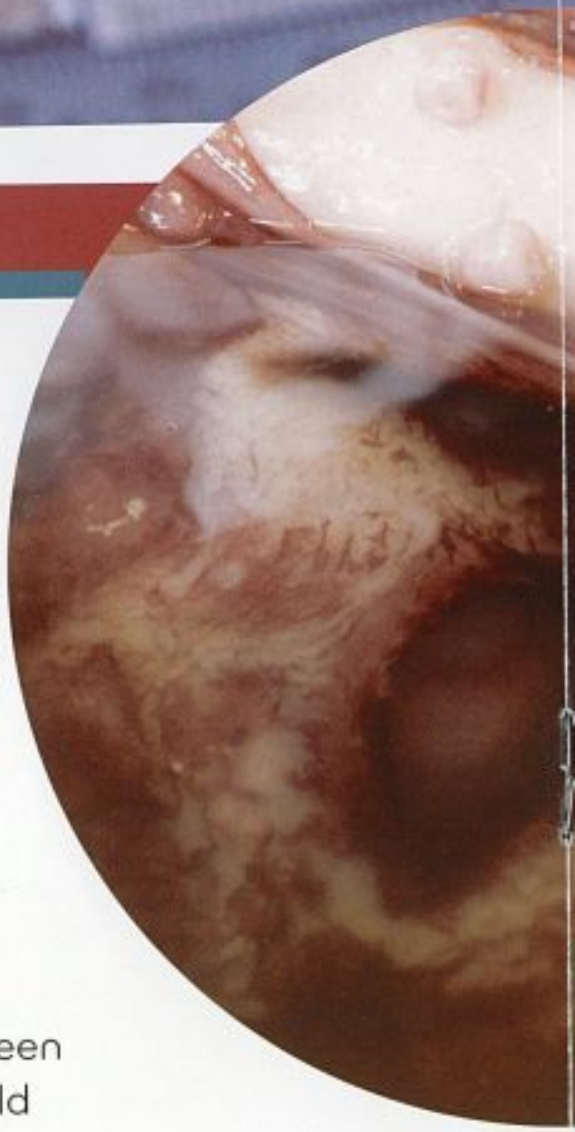
Floating plastic bags protecting the tentacles

Finally it is examination time. Scientists take measurements and tissue samples, count the suckers and claws, and use an underwater camera to work out how the squid is lying. The camera reveals the squid's enormous eye, which has a lens the size of an orange. Its eye doesn't have an eye socket and is just surrounded by skin, so it can easily pop out when caught by a hook or net or when it's under pressure as the squid is dragged up to the surface. This is the first time anyone has ever seen the eye of a colossal squid.

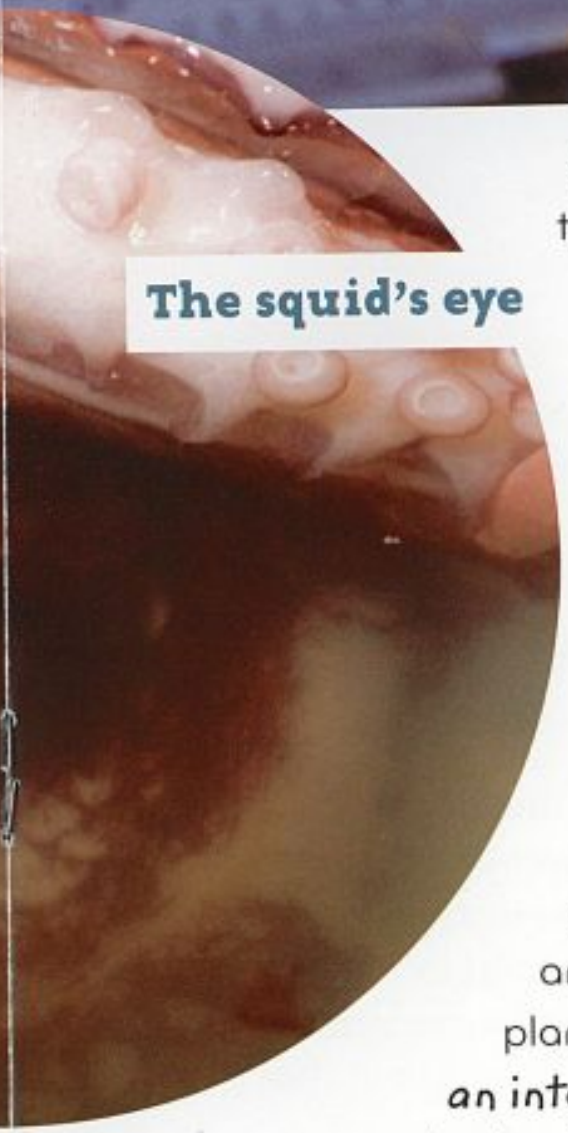
The whole world's watching

Te Papa staff set up a blog on their website so that people all round the world could follow what was happening. They installed a webcam and arranged for an audio link in the examination room.

By 30 April, there were up to 2000 people watching online at any one time, and the blog was receiving thousands of hits. People from 188 different countries were watching the colossal squid. Te Papa staff monitored how many people were accessing the website and where they came from. They could tell when a news item had been broadcast overseas because the website would immediately show a spike of interest from that country.



Measuring the squid



The squid's eye

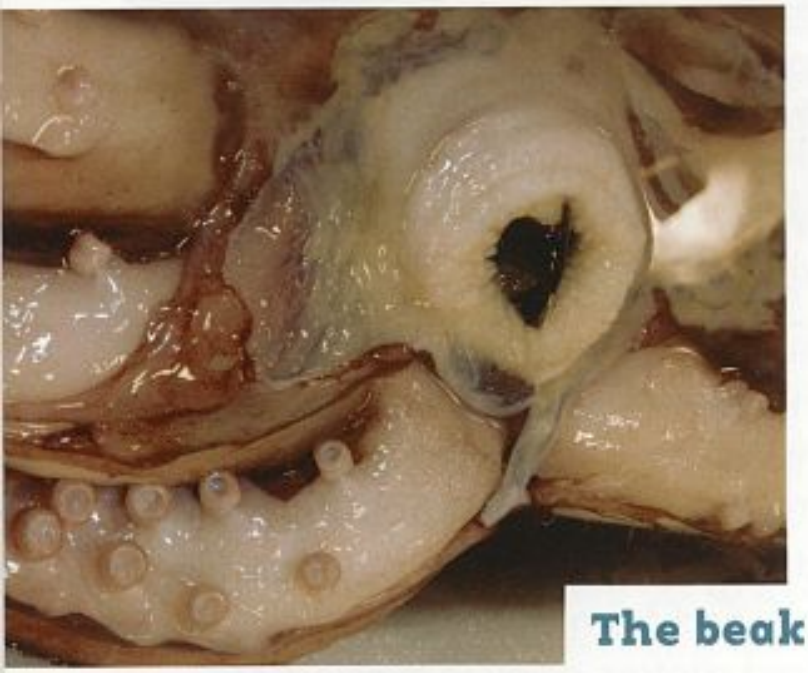
Newspapers and film crews were also covering the event. News items about the squid appeared on 64 000 different websites. It was the most viewed Internet story for the week.

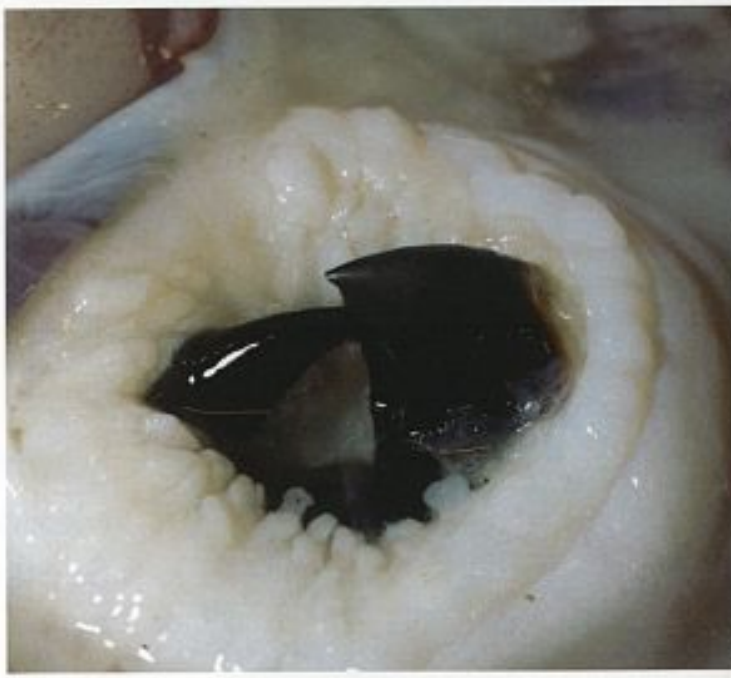
Updates were posted regularly on the blog. After each update, questions and comments poured in from people in India, Mexico, Germany, the United States, Scotland, England, Hungary, Singapore, Australia – and from people living only a few hundred metres away in Wellington. "*Fascinating!*" people wrote. "*I should be working, but I'm glued to the screen.*" Others were staying up all night to watch and be part of what was going on. Someone planned to hold a squid party. "*Thanks for such an interesting and unique experience. Thanks for sharing this wonder with the world,*" they wrote.

What the scientists discovered

- The squid measured 4.2 metres in length and weighed 495 kilograms.
- It had ovaries full of thousands of eggs, so it was a female.
- It was possibly 3–5 years old.
- Its last meal was Antarctic toothfish. Traces of toothfish eye lenses were in its stomach.

The fully thawed squid before being injected with formalin





The beak

- One of the most interesting discoveries was the length of the squid's beak. This colossal squid's lower beak measured 41.5 millimetres from its tip to its jaw. But other beaks as long as 49 millimetres have been found in the stomachs of sperm whales. This makes scientists wonder how much bigger colossal squid can grow. It's possible that there are colossal squid deep in the ocean that weigh as much as 700 or 750 kilograms.

Injecting with formalin

Fixing with formalin

Wednesday, 30 April

At 9 p.m., formalin is injected into the squid's tissues and pumped into the tank. This will preserve the tissue and make it harder – leathery, but not rigid. The pH levels are kept as neutral as possible so that the hooks and beak won't dissolve.

The final countdown

Wednesday, 6 August

The webcam is turned on again as the colossal squid is moved into its display tank. The formalin is drained off, and the squid is rinsed in fresh water. Scientists turn it over to repair rips and tears in the mantle. They use linen thread and as few stitches as possible. The eggs are taken out to be displayed separately. The squid is hoisted into a cradle, lifted, and placed in the new tank, and propylene-glycol preservative is pumped in.

It all takes about six hours. There are a few tense moments, but the blog reports: "*Our beautiful big squid is safely in her fancy tank and she looks great.*"

Monday, 17 November

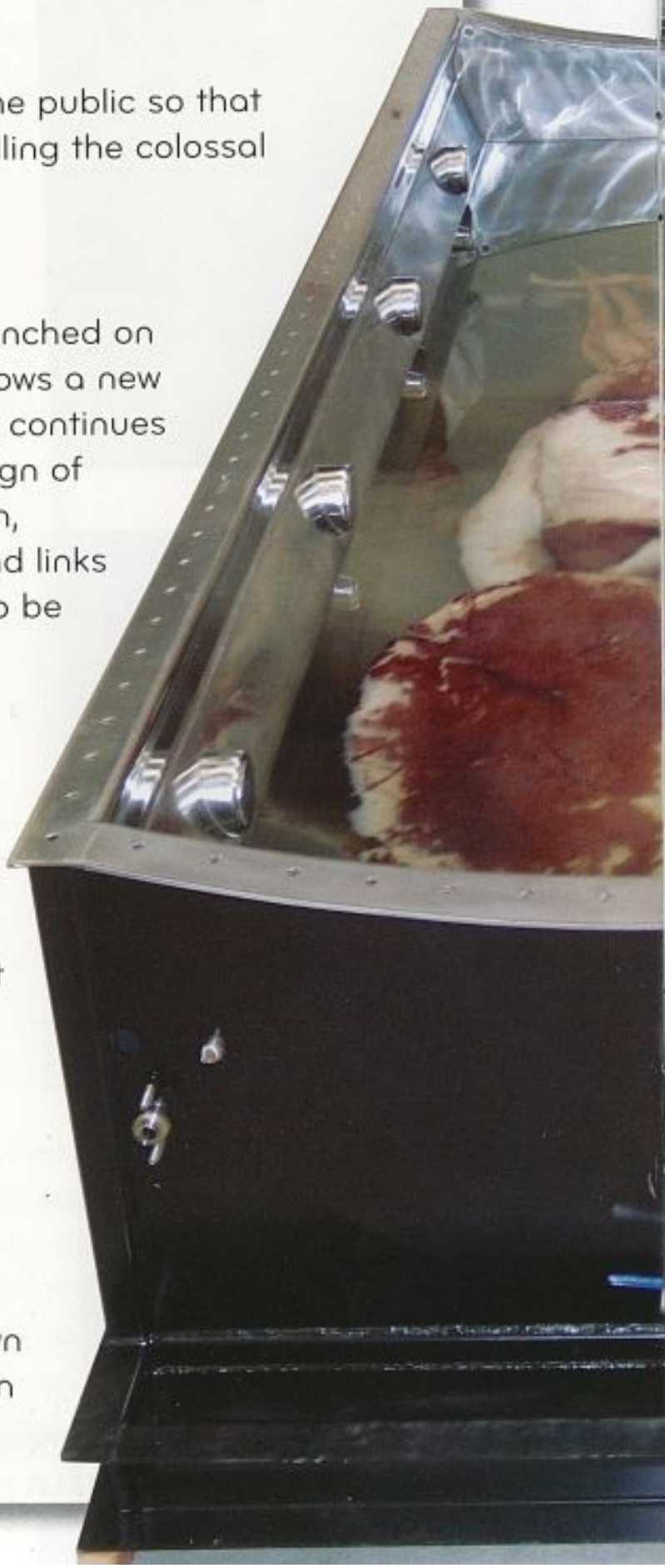
The museum gallery is closed to the public so that the exhibition team can start installing the colossal squid exhibition.

Thursday, 20 November

A countdown to opening day is launched on Te Papa's website. Every day, it shows a new image of the colossal squid. Work continues on the display panels and the design of the new squid website. Information, photographs, captions, buttons, and links all have to be checked. Text has to be finalised and labels printed.

Monday, 1 December

At 4 a.m., the lid is bolted onto the tank and the tank is hoisted onto a truck. The colossal squid is on the move! From the research laboratories in Tory Street to Te Papa is a distance of about five city blocks. It's important that the journey goes as smoothly as possible. A police escort stops traffic so the truck doesn't have to stop for red lights. The tank is unloaded at the goods entrance, and two forklift trucks guide it down corridors and nudge it into position in the gallery.



On the move to Te Papa

The exhibition at Te Papa

Saturday, 13 December

The colossal squid exhibition opens. Over 6000 people queue up to be among the first to see the only colossal squid ever on public display anywhere in the world.

The exhibition includes:

1. A screen playing a 5-minute video that shows the squid being caught and landed by the *San Aspiring*

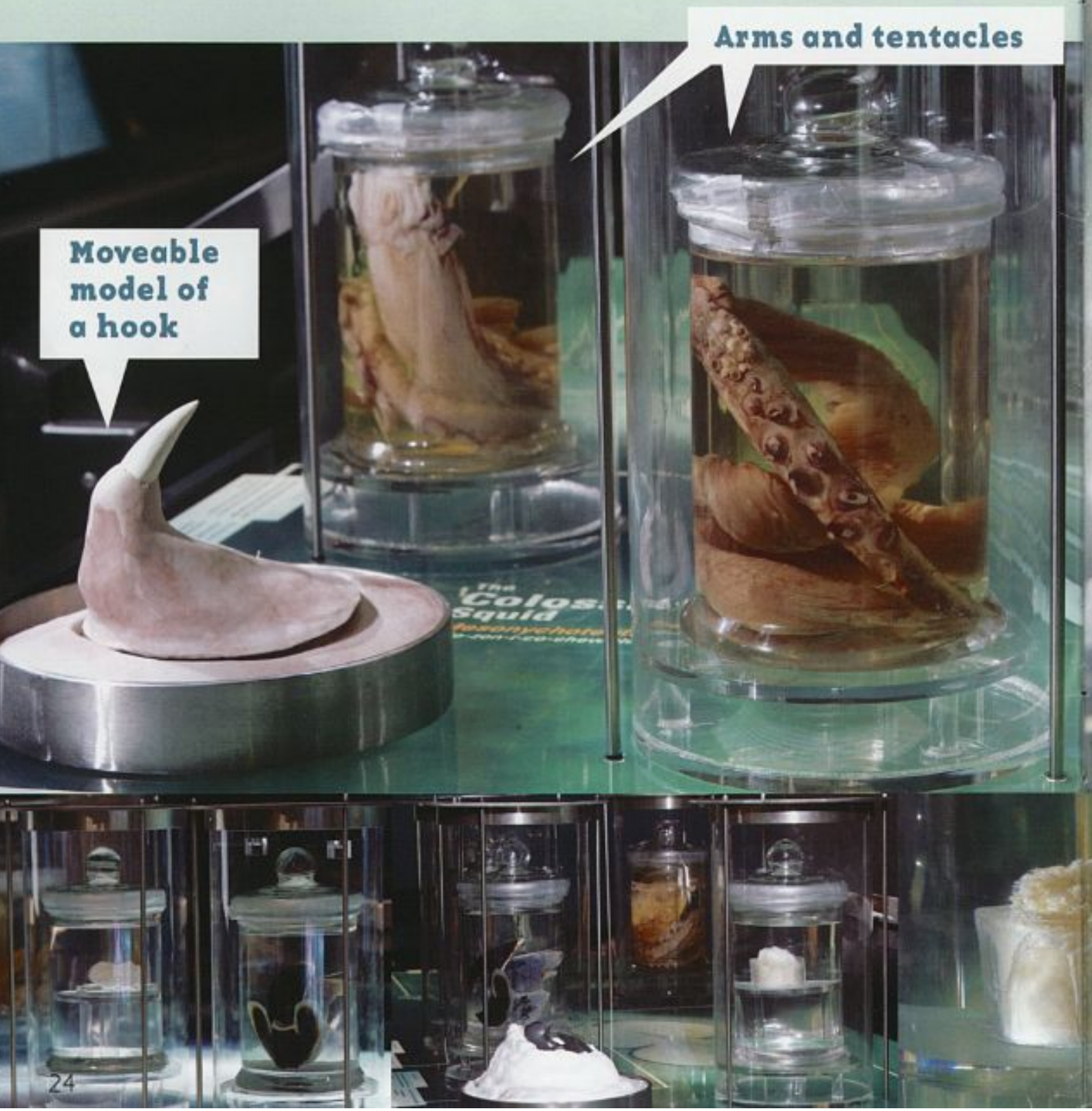
2. The tank:

- 4.6 metres long, 1.6 metres wide, 0.9 metres deep
- Weighs 2 tonnes (or 5 tonnes, including the liquid and squid), so it needs a floor strong enough to take its weight
- Made of stainless steel
- Lit from underneath to avoid any problem with surface reflections
- Curved acrylic lid to let air bubbles float to the side so they don't get in the way of viewers looking at the squid
- Trough to collect liquid if there is any spillage, for example, if there is an earthquake

- Acrylic supports (perspex rods) and cone-shaped mould to keep the squid in place and allow its tentacles to spread out
- Filled with 5000 litres of propylene-glycol preservative. Nobody has ever preserved such a large specimen in this way before. An alternative to formalin was needed because of the fire risk and regulations about use of dangerous chemicals in public spaces.
- Raised step so children can see inside
- Display panels at either end
- Interactive computer screens

3. Specimen and model displays:

- Specimen jars containing eggs, beak, eye lenses, arms and tentacles from smaller, dissected colossal squid
- Moveable models of squid beak and hooks
- Interactive computer screens



Beak

Model of the beak

Eggs

4. A 3-D movie: Created to show the colossal squid's way of life and its deep-sea environment. An animation company helped the scientists to make the squid look lifelike. Nobody knows for sure how the colossal squid swims, but the 3-D animation shows our best guess.